

Genetics and population analysis

Survival analysis on rare events using group-regularized multi-response Cox regression

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Abstract

Motivation: The prediction performance of Cox proportional hazard model suffers when there are only few uncensored events in the training data.

Results: We propose a Sparse-Group regularized Cox regression method to improve the prediction performance of large-scale and high-dimensional survival data with few observed events. Our approach is applicable when there is one or more other survival responses that 1. has a large number of observed events; 2. share a common set of associated predictors with the rare event response. This scenario is common in the UK Biobank dataset where records for a large number of common and less prevalent diseases of the same set of individuals are available. By analyzing these responses together, we hope to achieve higher prediction performance than when they are analyzed individually. To make this approach practical for large-scale data, we developed an accelerated proximal gradient optimization algorithm as well as a screening procedure inspired by Qian et al.

Availability and implementation: <https://github.com/rivas-lab/multisnpnet-Cox>

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1 Introduction

1.1 Cox proportional hazard model

Cox model (Cox, 1972) provides a flexible mathematical framework that describes the relationship between the predictors and a time-to-event response. For each individual, we observe a triple $\{O, X, T\}$, where $X \in \mathbb{R}^d$ are the features and $O \in \{0, 1\}$ is a status indicator. If $O = 1$, then T is the actual time-to-event. If $O = 0$, then we only know that the time-to-event is at least T . The hazard function according to the Cox model can be written as

$$h(t|X) = h_0(t) \exp(X^T \beta),$$

where $\beta \in \mathbb{R}^d$ is the coefficients vector that measures the strength of association between X and the response. This hazard function is equivalent to the cumulative distribution function:

$$P(T \leq t|X) = 1 - \exp\left(-\int_0^t h_0(s) e^{\beta^T X} ds\right).$$

Here $h_0: \mathbb{R}^+ \mapsto \mathbb{R}^+$ is the baseline hazard function. In our applications, we are interested in the relationship between the features

and the responses, so the baseline hazard function is a nuisance variable. We can estimate the parameters β directly without knowing the baseline hazard by maximizing the log partial likelihood function (Cox, 1972):

$$\begin{aligned} l(\beta|Data) &= \log \left[\prod_{i: O_i=1} \frac{\exp(X_i^T \beta)}{\sum_{j: T_j \geq T_i} \exp(X_j^T \beta)} \right] \\ &= \sum_{i=1}^n O_i [X_i^T \beta - \log \left(\sum_{j: T_j \geq T_i} \exp(X_j^T \beta) \right)] \end{aligned}$$

When the number of observed events is small relative to n , estimating β becomes challenging. This could happen, for example, when the time-to-event response is the age of diagnosis of a rare disease. In particular, if O_i are i.i.d Bernoulli random variables with probability p , then the information matrix is proportional to p and thus the asymptotic variance of the maximum partial likelihood estimate is inversely proportional to p .

We evaluate a fitted survival model using the concordance index, or the C-index. For a parameter estimate $\hat{\beta}$, the C-index $C(\hat{\beta})$ is defined as

$$\frac{\sum_{i=1}^n O_i |\{j : \hat{\beta}^T X_i > \hat{\beta}^T X_j\}| + |\{j : \hat{\beta}^T X_i = \hat{\beta}^T X_j\}|/2}{\sum_{i=1}^n O_i |\{j : T_j > T_i\}|}. \quad (1)$$

For more details on C-index, see Harrell *et al.* (1982) and Li *et al.* (2020).

1.2 Sparse-Group Lasso

The Lasso method (Tibshirani, 1996) makes the assumption that only a small subset of predictors are associated with the response. In other words, it assumes that β has only a small number of non-zero entries. A sparse solution can be obtained by optimizing an L_1 -regularized objective function.

Sparse-Group Lasso (Simon *et al.*, 2013) assumes not only that many individual elements of β are 0, but also that many groups of variables have coefficients 0 simultaneously. For example, in a single-response Cox model with d -dimensional features, if groups of variables $\mathcal{G} = \{g : g \subseteq [d]\}$ are believed to have sparse-group structure, then the sparse-group Lasso minimizes the following objective function:

$$-\sum_{i=1}^n O_i [X_i^T \beta - \log(\sum_{j: T_j \geq T_i} \exp(X_j^T \beta))] + \lambda \|\beta\|_1 + \sum_{g \in \mathcal{G}} \lambda_g \|\beta_g\|_2. \quad (2)$$

2 Approach

To improve the prediction performance of survival analysis on rare events, we pair each response with one or more survival responses with larger number of observed events. These responses are assumed to have a similar set of useful predictors as the rare event response. For each predictor, we group its coefficient on the multiple survival responses together and apply a sparse-group regularization on it to utilize the common support assumption.

3 Methods

3.1 Preliminaries

In this section, we define the notations and the key model assumptions that we will use in the subsequent sections. For an integer n , define $[n] = \{1, 2, \dots, n\}$ and define $x_+ = \max\{x, 0\}$ for all $x \in \mathbb{R}$.

We analyze $K \geq 1$ time-to-event responses on n individuals. For example, the responses could be time from birth to K different diseases. The data we observed are in the format:

$$\mathcal{D} = \{X_i, T_i^1, \dots, T_i^K, O_i^1, \dots, O_i^K\}_{i=1}^n.$$

Here $X_i \in \mathbb{R}^d$ are i th individual's features. Denote the full features matrix $X = [X_1, X_2, \dots, X_n]^T \in \mathbb{R}^{n \times d}$. For $k = 1, \dots, K$, $O_i^k = 1$ implies that T_i^k is the true time until the event k and for the i th individual, and $O_i^k = 0$ implies that the true time until the event k is right-censored at T_i^k . We assume each response follows a Cox proportional hazard model:

$$h_k(t|X) = h_{0,k}(t) \exp(X^T \beta_k). \quad (3)$$

where $h_{0,k} : \mathbb{R}^+ \mapsto \mathbb{R}^+$ is the baseline hazard function of the k th response. Let

$$n_k = \sum_{i=1}^n O_i^k$$

be the number of observed event k .

We make the assumption that not only β_k is sparse for all $k \in [K]$, but they also have a small common support. That is $\{j \in [d] : \beta_j^k \neq 0, k \in [K]\}$ is a small set relative to d . In human genetics applications, the first assumption means that each response is associated with a small set of genetic variants, and the second

assumption implies that there are significant overlap among the genetic variants that are associated with each response. This belief is the main driver for prediction performance improvements on less prevalent diseases, and it translates to the following regularized partial likelihood objective function:

$$\min_{\beta_1, \dots, \beta_K} \sum_{k=1}^K \frac{1}{n_k} \left[\sum_{i: O_i^k=1} -\beta_k^T X_i + \log \left(\sum_{j: T_j^k \geq T_i^k} \exp(\beta_k^T X_j) \right) \right] + \lambda \left(\sum_{j=1}^d \|\beta_j'\|_1 + \alpha \|\beta'\|_2 \right). \quad (4)$$

Here the first term is the sum of the K negative log-partial-likelihood, normalized by the number of observed events. The second term is the regularization. $\|\beta_j'\|_1$, $\|\beta'\|_2$ are the 1-norm and 2-norm of the coefficients for the j th variable. That is, if we put all the parameters into a matrix $B = [\beta_1, \beta_2, \dots, \beta_K] \in \mathbb{R}^{d \times K}$, then β_j' is the j th row of B and β_k is the k th column. Note that when $\alpha = 0$, the objective function decouples for each β_k and they can be optimized separately. In our implementation we allow each predictor to have different penalty factor, which may be useful in the setting where protein-truncating or protein-altering variants should be prioritized (Rivas *et al.*, 2015; DeBoever *et al.*, 2018). Just like in Yuan and Lin (2006), our implementation by default fixes α at $\sqrt{K}$. The solution is computed on a pre-defined sequence of λ s: $\lambda_1 > \lambda_2 > \dots > \lambda_L$, where λ_1 is chosen so that the solution just become non-zero.

To simplify the notation, for $j \in [d]$ denote

$$g_j = g_j(\beta_1, \dots, \beta_K) := \frac{\partial}{\partial \beta_j'} \sum_{k=1}^K \frac{1}{n_k} \left[\sum_{i: O_i^k=1} -\beta_k^T X_i + \log \left(\sum_{j: T_j^k \geq T_i^k} \exp(\beta_k^T X_j) \right) \right]. \quad (5)$$

Here $g_j \in \mathbb{R}^K$ is the partial derivative of the smooth part of (4) with respect to the coefficients of the variable j . Finally, let $S_1(\cdot; \lambda) : \mathbb{R}^K \mapsto \mathbb{R}^K$ be the element-wise soft-thresholding function. That is, for a vector $v \in \mathbb{R}^K$ with v_k its k -th coordinate, define:

$$S_1(v; \lambda)_k = \text{sign}(v_k)(|v_k| - \lambda)_+ \text{ for all } k \in [K].$$

3.2 Optimization method

We use a Nesterov-accelerated (Nesterov, 1983) proximal gradient method (Daubechies *et al.*, 2004; Beck and Teboulle, 2009) to optimize the objective function (4). Proximal gradient algorithm is particularly suitable when the objective function is the sum of a smooth function and a simple function. In our case, the smooth function is the sum of the negative log-partial-likelihood functions, and the simple function is the regularization term. The algorithm alternates between two steps until convergence criteria is met:

1. A gradient step that decreases the smooth part of the objective:
2. $\beta^j \leftarrow \beta^j - t g_j$, where t is the step size that we determine using backtracking line search.
3. A proximal step that keeps the regularization term small:
4. $\beta^j \leftarrow \text{prox}_t(\beta^j)$.
5. Here the proximal operator $\text{prox}_t : \mathbb{R}^K \mapsto \mathbb{R}^K$ is defined as
6. $\text{prox}_t(x) := \arg\min_x \frac{1}{2t} \|x - x\|_2^2 + \lambda \|x\|_1 + \alpha \|x\|_2$.
7. To simplify the notation, we omit the dependency of $\text{prox}_t(x)$ on λ, α . Simple calculation shows that
8. $\text{prox}_t(x) = S_2(S_1(x; t\lambda); t\alpha\lambda)$.

where $S_2(\cdot; t\alpha\lambda) : \mathbb{R}^K \mapsto \mathbb{R}^K$ is a group soft-thresholding operator:

$$S_2(v; t\alpha\lambda) = \begin{cases} 0 & \text{if } \|v\|_2 \leq t\alpha\lambda \\ \frac{v}{\|v\|_2} (\|v\|_2 - t\alpha\lambda) & \text{otherwise.} \end{cases}$$

We describe the optimization algorithm in pseudocode, including details about Nesterov acceleration and backtracking line search in Algorithm 1.

Algorithm 1: Proximal Gradient Method for (4)

Set line search parameter $\eta > 1$;
 Denote the parameter matrix $B \in \mathbb{R}^{d \times K}$. Initialize $B^{(0)}$;
 Write the smooth part of the objective as:

$$f(B) = \sum_{k=1}^K \frac{1}{n_k} \left[\sum_{i: O_i^k=1} -\beta_k^T X_i + \log \left(\sum_{j: T_j^k \geq T_i^k} \exp(\beta_k^T X_j) \right) \right].$$

Set iteration count $i = 0$; Set initial step-size $t = 1$; Set Nesterov weights $w_0, w_1 = 1$;

while B has not converged **do**

Nesterov acceleration: $w_1 \leftarrow (1 + \sqrt{1 + 4w_0^2})/2$;
 $B^{(i+0.5)} \leftarrow B^{(i)} + (w_0 - 1)(B^{(i)} - B^{(i-1)})/w_1$;
 $w_0 \leftarrow w_1$;
 Compute the gradient $g_j = \frac{\partial}{\partial \beta^j} f(B^{(i+0.5)})$ for $j \in [d]$;
 Start backtracking line search:
repeat
 $B \leftarrow B^{(i+0.5)} - t \nabla f(B^{(i+0.5)})$;
 Denote β^j the j th row of B for $j \in [d]$;
 Apply proximal step: $\beta^j \leftarrow \text{prox}_t(\beta^j)$ for $j \in [d]$;
 if $f(B) \leq f(B^{(i+0.5)}) + \langle B - B^{(i+0.5)}, \nabla f(B^{(i+0.5)}) \rangle + \|B - B^{(i+0.5)}\|_2^2 / (2t)$ **then**
 break;
 $t \leftarrow t/\eta$;
until the break condition above is satisfied ;
 $B^{(i+1)} \leftarrow B$; $i \leftarrow i + 1$;
 Check convergence based on objective value change or parameter change.

return $B^{(i)}$

3.3 Variable screening for Lasso Path

In many of our applications, the data are large-scale and high-dimensional. For example, the UK Biobank dataset (Sudlow *et al.*, 2015) contains millions of genetic variants and over 500 000 participants. Reading the feature matrix of the UK Biobank dataset into R requires more than 4 terabytes of memory, which is much larger than the RAM size of most computers. While memory mapping (Kane *et al.*, 2013) allows users to access out-of-memory data with ease, it requires lots of disk Input/Output operations, which is much slower than in-memory operations. This becomes even more problematic for iterative optimization algorithms that use the entire feature matrix every iteration.

To reduce the frequency of reading the full data matrix, here we derive a version of variable screening method following similar ideas of the strong rule (Tibshirani *et al.*, 2012) and the Batch Screening Iterative Lasso (Qian *et al.*, 2020).

For each $\alpha > 0, v \in \mathbb{R}^K$, define the α^* norm of v to be

$$\|v\|_{\alpha^*} := \sup\{u^T v : u \in \mathbb{R}^K, \|u\|_1 + \alpha \|u\|_2 \leq 1\}. \quad (6)$$

We use the following results to get the optimality conditions, or the Karush–Kuhn–Tucker (KKT) conditions, for the solutions $\beta_1, \dots, \beta_K$. The proofs are given in Section 7.

Proposition 1 For any $\lambda > 0$, the gradients defined at (5) at the optimal solution to (4) satisfies:

$$\|g_j\|_{\alpha^*} \begin{cases} \leq \lambda & \text{if the optimal } \beta^j = 0 \\ = \lambda & \text{if the optimal } \beta^j \neq 0 \end{cases} \text{ for all } j \in [d]. \quad (7)$$

This result motivates us to first fit a model (solving (4)) using a small number of variables whose gradient has the largest α^* norm, assuming the coefficients for the rest of the variables are all zero. Then to verify the validity of the assumption we check that $\|g_j\|_{\alpha^*} \leq \lambda$ for variables assumed to have zero coefficients. We refer to this

step as KKT checking. If KKT checking succeeds, then we obtain a valid solution to (4). Otherwise, we increase the number of variables used in the fitting step based on their α^* norm until KKT checking succeeds. As a result, this screening method always gives the exact solution to the original optimization problem. Note that based on its definition (6), it's not clear how we can compute $\|v\|_{\alpha^*}$. Here we give a more explicit characterization.

Proposition 2 $\|v\|_{\alpha^*} \leq \lambda$ if and only if $\|S_1(v; \lambda)\|_2 \leq \alpha \lambda$.

Using the above we can check if $\|v\|_{\alpha^*} \leq \lambda$ quite easily and compute $\|v\|_{\alpha^*}$ using binary search in $\mathcal{O}(K \log \|v\|_{\infty})$ to any fixed precision (since $\|v\|_{\alpha^*} \leq \|v\|_{\infty}$).

Now we are ready to state the overall structure of our algorithm with variable screening. Suppose valid solutions for $\lambda_1, \dots, \lambda_l$ have been obtained. Next, we follow these steps:

1. **(Screening)** In the last iteration, we cache the full gradient $\{g^j\}_{j=1}^d$ evaluated at the solution at λ_l . In the fitting step we include two types of variables:
 - We include the ever-active variables $\mathcal{A} := \{j \in [d] : \hat{\beta}^j \neq 0 \text{ for any previously obtained } \hat{\beta}\}$.
 - Top M variables with the largest $\|g_j\|_{\alpha^*}$ that are also not ever-active.
 - We denote the set of variables used to fit (4) as the strong set $\mathcal{S} \subseteq [d]$.
1. **(Fitting)** In this step, we solve the problem (4) for the next few λ s using only variables in $\mathcal{S}$, assuming $\beta^j = 0$ for all $j \notin \mathcal{S}$. This is done using proximal gradient descent (Algorithm 1). To speed-up the computation we initialize the variables at the previous valid solution (warm start).
2. **(KKT Checking)** After obtaining the solution from the fitting step. We compute the full gradient. This is the only step we will need the full data matrix X . We check if the KKT conditions (7) are satisfied for all variables then go back to the screening step at the first λ value where the KKT condition fails. We also cache the full gradient at the last valid solutions for the screening step.
3. **(Early Stopping)** We keep a separate validation set to evaluate the current estimated parameters. We choose the optimal λ as the one that gives the highest validation C-index. The optimal λ might be different for different responses. In this paper we focus only on the prediction accuracy of one rare event, so it is reasonable to stop when the validation C-index of this response starts to decrease, regardless if the optimal λ for other responses has been reached.

These steps are described in Algorithm 2.

3.4 Software

We implemented the proximal gradient method in Section 2.1 as an R package, available at <https://github.com/rivas-lab/multisnpnet-Cox>. In this package we also implement the screening procedure described in this section for genetics data in Plink2 format (Chang *et al.*, 2014).

The major computational bottleneck in the proximal gradient method are matrix-matrix multiplications. These operations have high arithmetic intensity and are particularly suitable for GPU acceleration. Therefore, we also provide a GPU implementation of Algorithm 1 on CUDA-enabled devices, available at https://github.com/rivas-lab/multisnpnet-Cox_gpu. With $n = d = 10\,000, K = 20$, the GPU implementation achieves almost $10\times$ speedup in solving a path of 50 λ s (9.7 seconds versus 92 seconds) on a Tesla V100 GPU than the CPU implementation on an Intel Xeon 6528R processor with 28 cores, accelerated using Intel's Math Kernel Library.

Algorithm 2: Iterative Screening for Lasso Path

Initialize ever active set $\mathcal{A}^{(0)} = \emptyset$ and construct the regularization parameters $\lambda_1, \dots, \lambda_L$;
Initialize a short list of initial regularization parameters
 $\Lambda^{(0)} = \{\lambda_1, \dots, \lambda_{L^{(0)}}\}$;
Initialize the parameters $\hat{\beta}_1^{(0)}, \dots, \hat{\beta}_K^{(0)} = 0$; Set the iteration counter $i = 0$;
while $\hat{\beta}(\lambda_L)$ not computed **do**
 Set v to be the largest number such that the solutions for $\lambda_1, \dots, \lambda_v$ are obtained;
 Screening:
 Compute (or use the cached) full gradient (5) $g_1, \dots, g_d$ at the last solutions.
 Set $\mathcal{E}_M^{(i)}$ to be the M variables in $[d] \setminus \mathcal{A}^{(i)}$ with the largest $\|g_j\|_{\alpha*}$;
 Set $\mathcal{S}^{(i)} = \mathcal{A}^{(i)} \cup \mathcal{E}_M^{(i)}$. This will be the variables used in the fitting step;
 Fitting:
 for $\lambda \in \Lambda^{(i)}$ **do**
 Obtain parameter estimates by solving (4) using only variables in $\mathcal{S}^{(i)}$;
 The optimization algorithm is described in algorithm 2;
 Coefficients for variables not in $\mathcal{S}^{(i)}$ are set to be zero.;
 Checking:
 Compute the full gradients $g_1(\lambda), \dots, g_d(\lambda)$ at solutions obtained with regularization parameters $\lambda \in \Lambda^{(i)}$;
 Find the smallest $\lambda \in \Lambda^{(i)}$ such that the KKT condition (7) is satisfied:

$$\bar{\lambda}^{(i)} = \min \left\{ \lambda \in \Lambda^{(i)} : \max_{j \in [d] \setminus \mathcal{S}^{(i)}} \|g_j(\lambda)\|_{\alpha*} < \lambda \right\}$$

 Update ever active set $\mathcal{A}^{(i+1)} = \mathcal{A}^{(i)} \cup \{j : \hat{\beta}^j(\bar{\lambda}^{(i)}) \neq 0\}$;
 Update $\Lambda^{(i+1)} = \{\lambda \in \Lambda^{(i)} : \lambda < \bar{\lambda}^{(i)}\}$. Extend $\Lambda^{(i+1)}$ if it is too short.;
 For $\lambda \in \{\lambda \in \Lambda^{(i)} : \lambda \geq \bar{\lambda}^{(i)}\}$, we obtain valid solutions; Set $i = i + 1$;
return $\hat{\beta}(\lambda_1), \dots, \hat{\beta}(\lambda_L)$

4 Simulations

In this section, we compare the performance of the proposed approach against a simple Lasso, where multiple responses are fitted independently. Here we simulate two responses ($K=2$), $n=400$, $d=5000$, the entries of the predictor matrix are i.i.d random signs $\{-1, 1\}$ with probability 0.5 each, and the time-to-event responses are exponential distributed with rate $\exp(X_i^T \beta)$, which satisfies proportional hazard. The parameters β_1, β_2 have a common support of size 35. We use a large (5000 samples) and uncensored validation set to select the optimal λ . α is fixed at $\sqrt{2}$. The parameter estimate corresponding to the best λ is then evaluated at a large, uncensored test set. We use the C-index as both the validation and test metric. The censoring for both responses are randomly chosen and independent from everything else. The simulations are done for multiple combinations of censoring proportions, each repeated 100 times. Here we report the improvement in C-index of the response with rare events. See Figures 1 and 2.

The assumption that the coefficients for different responses have the same support can come from domain knowledge (such as biology). In practice, this is usually not exactly satisfied. Here we use simulation to examine the robustness of our approach. The setup is the same as the previous simulations, except now the overlap of the support might be smaller than 35 (the number of non-zero entries of β_1, β_2). See the left panel of Figure 3.

We can see that, when the support overlap percentage is less than around 40% the prediction performance actually becomes worse when the two responses are trained together. One solution to this problem is to also treat α as a hyperparameter and use the validation set to determine it. For large data set having a two-dimensional hyperparameter could be quite cumbersome. In our simulation and real data application, we only choose α from two values $\{0, \sqrt{K}\}$, although in principle one can use a large set of α candidates at a higher computational cost. The right panel of Figure 3 shows the C-index improvements when we use a small validation set to determine whether α should be 0 or $\sqrt{K}$. All other settings are the same as above. However, tuning α using a validation set with few observed events also increases the variance of test C-index.

5 Application to UK Biobank data

We focus on thyroiditis, which has 808 observed events in the study population (337, 129 unrelated participants of white British ancestry, as described in DeBoever *et al.* (2018)). We randomly assign 70% of the samples to the training set, 10% to the validation set, and 20% to the test set. Focusing on the entire study population of 337, 129 individuals and 147, 604 variants that are marked as ‘in_PCA’ in the variant QC file (ukb_snp_qc.txt) from UK Biobank Sudlow *et al.* (2015), we compute the principal components of genetic variants using ‘pca approx’ sub-command in PLINK2 (Chang *et al.*, 2014). We use the top 10 principal components and sex as covariates. For genetic features, we first combine the directly genotyped variants (release version 2 of Sudlow *et al.* (2015)), the imputed allelotypes in human leukocyte antigen allelotypes (Venkataraman *et al.*, 2020), and copy number variations described in Aguirre *et al.* (2019), resulting in a genotype matrix of 1 080 968 variants, as described in Sinnott-Armstrong *et al.* (2021). As a prepossessing step, we exclude variants with missing rate greater than 10% and variants whose minor allele frequency is less than 0.001, which leaves approximately 700 000 markers remaining as predictors. We first fit a baseline model using only the 11 covariates, without using any genetic variants. This gives a baseline test C-index at 0.649. We then fit a single-response Lasso Cox regression, where λ is selected using based on the validation C-index. At the best λ value, the test C-index is 0.679. To apply our approach, we pair thyroiditis with 6 other more common endocrine diseases that we believe share common genetic factors with thyroiditis. These diseases are listed in the table below. Here we report the test C-indices when thyroiditis is paired with one other disease and when all 7 responses are trained together. We also report the test C-indices when we use the validation set to determine both the optimal λ and whether to set $\alpha = 0$ or $\alpha = \sqrt{K}$.

Table 1 shows clear increase in prediction performance on thyroiditis when all 7 responses are trained together, and when thyroiditis is paired with thyrotoxicosis only. On the other hand, all multi-response solutions have test C-index comparable with the single response solution. It is worth noting that selecting α using the validation set might not always give better test C-index than using a fixed α . In the case of other non-toxic goitre, tuning α gives slightly worse test metric due to the increased variance incurred in this procedure.

In terms of computation, the overall time duration and memory requirements depend mostly on the number of variables selected by the screening procedure (size of $\mathcal{S}$ in Algorithm 2) that we use to fit the last few models before early stopping happens. In all of the scenario above, 1500–2000 variables are used for the fitting step in the end. For a fixed α each case took around 10–18 minutes to finish on an Intel Xeon 6528 R processor using 16 cores and 230GB of RAM. When fitting is done on a Tesla V100 GPU, the time duration becomes 6–18 minutes with comparable prediction performance.

6 Discussion

We developed a regularized regression method for multiple survival responses. This method is particularly suitable when we can pair a

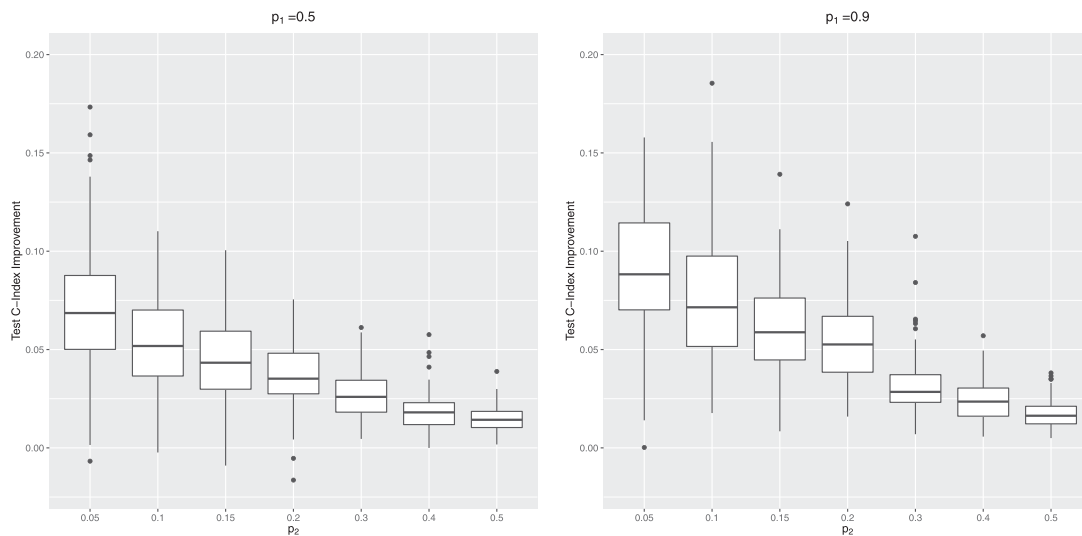


Fig. 1. Absolute improvements in test C-index of the rare event response, as p_2 varies. p_1 is the proportion of uncensored events of the first response. p_2 (rare) is the proportion of uncensored events of the second response. The true coefficients have exactly the same support.

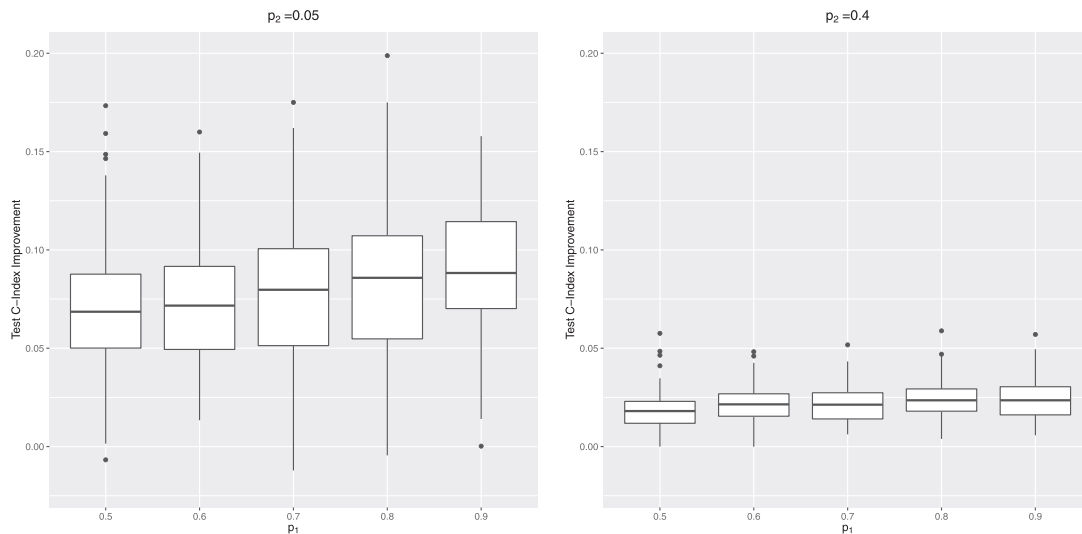


Fig. 2. Absolute improvements in test C-index of the rare event response, as p_1 varies. In contrast to Figure 1, each panel here has a fixed proportion of uncensored events in the second (rare event) response. The horizontal axis is the proportion of uncensored events of the first response (p_1). The true coefficients have exactly the same support.

response with few observed events with others having a larger number of events, such that these responses have a same set of useful predictors. We demonstrate the improvements in the prediction accuracy for the rare events responses through simulation and real data applications. While we believe our method will be a useful tool in practice, we do not give theoretical properties of the estimators obtained by solving (4). On the practical side, it is reasonable to have different types of models (not just Cox model), or even different response types (such as binary or count) to boost the accuracy of survival analysis on rare events. In principle the same type of regularization could still be applied. The prediction performance under this situation is yet to be explored.

For the GPU implementation, one challenge is the limited amount of memory available on GPUs. In modern computer clusters it is common to have machines with hundreds of gigabytes of system memory, but most GPUs have less than 32GB of memory. In our implementation we use single precision floating point numbers to store the data matrix, which alleviates memory burden by a factor of two. However, for UK Biobank scale data this is still sometimes insufficient. For example, when the number of participants in the training

data is 250 000, we are only able to fit a model with up to 32 000 variables on a Tesla V100. One possible solution is to use multiple GPUs, but inter-GPU communication might become the bottleneck. For genetics data, another solution is to utilize their 2-bit representation, which can significantly reduce memory requirement (2 bits per entry versus 32). We leave these ideas for future studies.

7 Proof of propositions

7.1 Proof of Proposition 1

We show a slightly more general result from convex analysis. Let $f : \mathbb{R}^K \rightarrow \mathbb{R}$ be continuously differentiable, convex, and bounded from below. Let $\|\cdot\|$ be a norm on $\mathbb{R}^K$, and $\|\cdot\|_*$ be its corresponding dual norm. Let $\lambda > 0$, and set

$$x^* := \operatorname{argmin}_x f(x) + \lambda \|x\| \quad (8)$$

We will show that

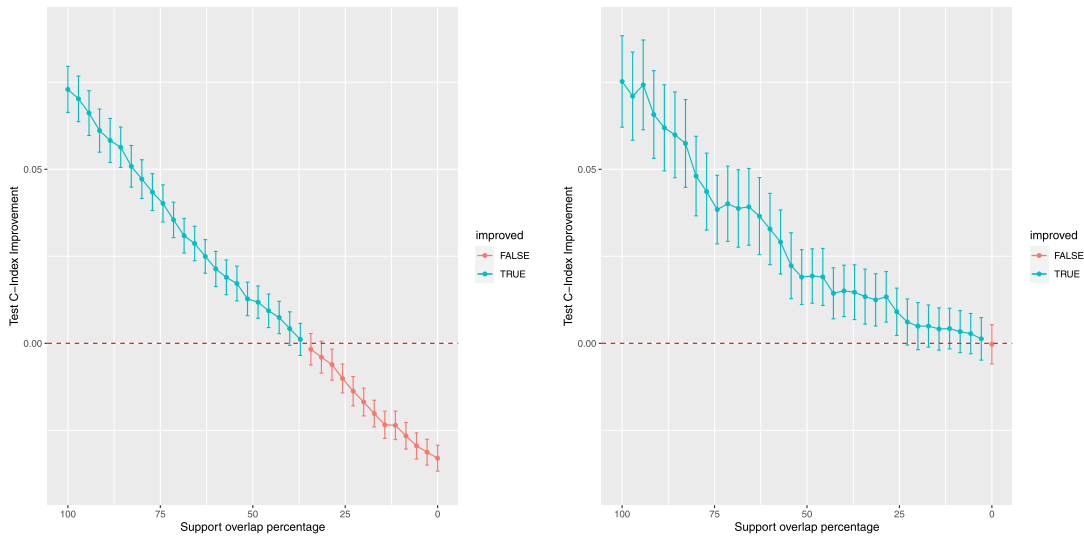


Fig. 3. Absolute improvements in test C-index of the rare event response as support overlap varies. 60% of the events for the first response are uncensored, and 5% of the events for the second are uncensored. The horizontal axis is the overlap proportion of the support of β_1 and β_2 , in other words $|\{i \in [p] : \beta_1^i \neq 0, \beta_2^i \neq 0\}| \cdot 100/35$. The left panel shows the result when α is fixed at $\sqrt{2}$, and λ is selected from a large uncensored validation set. The right panel shows the result when α is selected between $\{0, \sqrt{2}\}$ using a small validation set of size 200 with 5% uncensored second event. The whiskers indicates 95% confidence intervals.

Table 1. Test C-index of thyroiditis when this response is paired with other ones. The second column is obtained when α is fixed at $\sqrt{K}$ (which is 1 in the first row, $\sqrt{7}$ in the last row, and $\sqrt{2}$ in the rest). The third column is obtained when we use the validation set to choose α between 0 (single-response) and $\sqrt{K}$. Improved C-indices are given in boldface. Baseline test C-index of thyroiditis, when only the 11 covariates are used for fitting, is 0.649.

Paired response(s)	Test C	Test C (validate α)	Events Count
Thyroiditis (single response)	0.679	—	808
Other hypothyroidism	0.682	0.682	22829
Other non-toxic goitre	0.681	0.679	3011
Thyrotoxicosis (hyperthyroidism)	0.686	0.686	4705
Other disorders of thyroid	0.679	0.679	1415
Insulin-dependent diabetes mellitus	0.683	0.683	2782
Non-insulin-dependent diabetes mellitus	0.679	0.679	20549
All 7 responses trained together	0.688	0.688	—

$$\|\nabla f(x^*)\|_* \leq \begin{cases} \lambda & \text{if } x^* = 0 \\ \lambda & \text{if } x^* \neq 0 \end{cases} \quad (9)$$

It is clear that (8) is equivalent to the constrained optimization problem

$$\min_{x,y} f(x) + \lambda \|y\| \text{ such that } x = y. \quad (10)$$

The Lagrangian is

$$\ell(x, y, z) = f(x) + \lambda \|y\| + z^T(y - x). \quad (11)$$

The Lagrangian dual is

$$g(z) := \inf_{x,y} \ell(x, y, z) = \inf_y \lambda \|y\| + z^T y + \inf_x f(x) - z^T x. \quad (12)$$

Using the definition of dual norm, when $\|z\|_* > \lambda$, the infimum of the first term above is $-\infty$, and when $\|z\|_* \leq \lambda$, the infimum is 0. Therefore

$$g(z) = \begin{cases} -\infty & \text{if } \|z\|_* > \lambda \\ \inf_x f(x) - z^T x & \text{if } \|z\|_* \leq \lambda \end{cases} \quad (13)$$

Therefore, the dual solution $z^* := \operatorname{argmax}_z g(z)$ must satisfy $\|z\|_* \leq \lambda$. Now we go back to the Lagrangian. Since the primal objective is convex and Slater's condition holds, the solution to the primal problem can be obtained through minimizing

$$\ell(x, y, z^*) = f(x) + \lambda \|y\| + z^{*T}(y - x). \quad (14)$$

which implies that, at the optimal x^* we must have $\nabla f(x^*) = z^*$, so

$$\|\nabla f(x^*)\|_* = \|z^*\|_* \leq \lambda, \text{ and } \lambda \|x^*\| + \nabla f(x^*)^T x^* = 0 \quad (15)$$

If $x^* = 0$, then the second equality is already satisfied, so we only need $\|\nabla f(x^*)\|_* \leq \lambda$. If $x^* \neq 0$, then by Holder's inequality

$$\lambda \|x^*\| = |\nabla f(x^*)^T x^*| \leq \|\nabla f(x^*)\|_* \|x^*\| \leq \lambda \|x^*\|, \quad (16)$$

so we must have $\|\nabla f(x^*)\|_* = \lambda$. Proposition 1 is a direct consequence of this result.

7.2 Proof of Proposition 2

We prove the claim for $\lambda = 1$. The regularization term can be written as

$$\|u\|_1 + \alpha \|u\|_2 = \sup_{\|v_1\|_\infty \leq 1} v_1^T u + \sup_{\|v_2\|_2 \leq \alpha} v_2^T u = \sup_{v \in \mathcal{B}_\alpha^*} v^T u \quad (17)$$

where $\mathcal{B}_\alpha^*$ is the unit dual ball

$$\mathcal{B}_\alpha^* := \{v_1 \in \mathbb{R}^K : \|v_1\|_\infty \leq 1\} \oplus \{v_2 \in \mathbb{R}^K : \|v_2\|_2 \leq \alpha\}. \quad (18)$$

That is $\|v\|_{\alpha^*} \leq 1$ if and only if $v \in \mathcal{B}_{\alpha^*}$, which by definition means that $v = v_1 + v_2$ for some $\|v_1\|_{\infty} \leq 1$, $\|v_2\|_2 \leq \alpha$. We must have (and it is sufficient to have)

$$\inf_{\|v_1\|_{\infty} < 1} \|v - v_1\|_2 \leq \alpha. \quad (19)$$

The infimum on the left-hand side is achieved when $v_1 = v - S_1(v, 1)$, which proves the claim.

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Conflict of Interest:

The authors do not declare any conflict of interest.

Data availability

The data that support the findings of this study are available for approved research. Access to the data can be applied through UK Biobank's website <https://www.ukbiobank.ac.uk/enable-your-research/apply-for-access>

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